
Algorithm: Constraint and Consistency of Synthetic Images in BiomedCLIP Space

Require: Real image folders $\mathcal{R}_{\text{mild}}, \mathcal{R}_{\text{severe}}$
Require: Synthetic folders $\{\mathcal{S}_k\}_{k=1}^K$ (each labeled mild/severe)
Require: BiomedCLIP vision encoder $f(\cdot)$ and preprocessor $g(\cdot)$
Require: Batch size B
Ensure: Table with columns: Generator, Severity, N , Mean_CosSim, Std_CosSim, Min, Max, Range, $\Delta\mu$, ρ ; optional one-way ANOVA (F, p) per severity

- 1: **function** EMBEDANDNORM($\mathcal{I}, f, g, B$)
- 2: Split $\mathcal{I}$ into batches of size B
- 3: **for each** batch $\mathcal{B}$ **do**
- 4: $F \leftarrow f(g(\mathcal{B}))$
- 5: Row-wise L2-normalize $F \leftarrow F/\|F\|_2$
- 6: Accumulate rows of F into list
- 7: **end for**
- 8: **return** concatenated, unit-normalized embeddings $\mathbf{Z} \in \mathbb{R}^{|\mathcal{I}| \times D}$
- 9: **end function**
- 10: $Z_{\text{mild}} \leftarrow \text{EMBEDANDNORM}(\mathcal{R}_{\text{mild}}, f, g, B)$
- 11: $Z_{\text{severe}} \leftarrow \text{EMBEDANDNORM}(\mathcal{R}_{\text{severe}}, f, g, B)$
- 12: $\mu_{\text{mild}} \leftarrow \text{Normalize}\left(\frac{1}{|Z_{\text{mild}}|} \sum_{z \in Z_{\text{mild}}} z\right)$
- 13: $\mu_{\text{severe}} \leftarrow \text{Normalize}\left(\frac{1}{|Z_{\text{severe}}|} \sum_{z \in Z_{\text{severe}}} z\right)$
- 14: Initialize empty results table $\mathcal{T}$
- 15: **for each** synthetic set $\mathcal{S}_k$ with label $c \in \{\text{mild}, \text{severe}\}$ **do**
- 16: $X_k \leftarrow \text{EMBEDANDNORM}(\mathcal{S}_k, f, g, B)$
- 17: $\mu_c \leftarrow \begin{cases} \mu_{\text{mild}}, & c = \text{mild} \\ \mu_{\text{severe}}, & c = \text{severe} \end{cases}$
- 18: $S_k \leftarrow \{s_j = x_j^\top \mu_c \mid x_j \in X_k\}$
- 19: $\bar{s}_k \leftarrow \frac{1}{|X_k|} \sum_j s_j$; $\sigma_k \leftarrow \sqrt{\frac{1}{|X_k|} \sum_j (s_j - \bar{s}_k)^2}$
- 20: $m_k \leftarrow \min S_k$; $M_k \leftarrow \max S_k$; $r_k \leftarrow M_k - m_k$
- 21: $\hat{\mu}_k^{(\text{syn})} \leftarrow \text{Normalize}\left(\frac{1}{|X_k|} \sum_{x \in X_k} x\right)$
- 22: $\Delta\mu_k \leftarrow 1 - \mu_c^\top \hat{\mu}_k^{(\text{syn})}$ // centroid misalignment (smaller is better)
- 23: $\rho_k \leftarrow \sigma_k$ // dispersion around μ_c
- 24: **Append** (Generator = k , Severity = c , $N = |X_k|$, Mean_CosSim = $\bar{s}_k$,
- 25: Std_CosSim = σ_k , Min = m_k , Max = M_k , Range = r_k , $\Delta\mu = \Delta\mu_k$, $\rho =$
- 26: ρ_k) to $\mathcal{T}$
- 27: **end for**
- 28: **for** $c \in \{\text{mild}, \text{severe}\}$ **do** ▷ Optional ANOVA
- 29: Gather $\{S_k : \text{Severity}(k) = c\}$ across generators and compute ANOVA (F, p)
- 30: Record (c, F, p) alongside $\mathcal{T}$
- 31: **end for**
- 32: **Return** $\mathcal{T}$ (and ANOVA results if computed)
